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1  SARKARI SAHAYAK – VOICE PIPELINE: OUTSTANDING ITEMS BEFORE/AFTER DEPLOYMENT
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3
4  BLOCKER – fix before any multi-caller usage, not safe to defer
5  -----
6  [ ] Per-call room isolation
7      room_name = "support-room" is hardcoded in transport.py / main.py /
8      voice_server.py's /token route. Every caller currently joins the SAME
9      LiveKit room. With more than one simultaneous caller, audio/responses
10     will mix across unrelated conversations. Needs a unique room per call
11     (UUID/session id), or LiveKit SIP dispatch rules configured to auto-
12     create a room per inbound call.
13
14
15  HIGH PRIORITY – real risk of breakage or bad user experience
16  -----
17  [ ] Real SIP/DTMF testing
18      The entire IVR language-lock design (ivr.py, languages.py) has only
19      been tested via browser/mic, where it falls back to the timeout ->
20      DEFAULT_DIGIT path. Real phone-line DTMF tones have never been
21      verified end to end. Needs an actual LiveKit SIP trunk + real call.
22
23  [ ] Cloudflare Workers AI quota
24      Already hit both the per-minute rate limit AND the 10k/day free
25      neuron cap during solo testing (embeddings.py). Real call volume
26      will exhaust this fast. Needs a paid tier / usage-based billing
27      before real traffic, not after.
28
29  [ ] /token has no auth
30      voice_server.py's /token route hands out a valid LiveKit join token
31      to anyone who calls it, with a hardcoded identity ("user-123"). Needs
32      to be gated appropriately for however the real call flow reaches it.
33
34  [ ] Only Hindi has been conversation-tested
35      languages.py declares English and Tamil as supported (KeypadEntry.TWO,
36      THREE) but neither has been exercised through a real conversation yet
37      – only Hindi has confirmed working end to end (IVR lock, LLM reply
38      language, TTS/STT retuning).
39
40  [ ] Retrieval quality after removing hard filters
41      state/government_level filters were dropped from tool.py / retriever.py
42      in favor of folding that context into the query text (soft semantic
43      signal instead of hard filter). Validated on a couple of examples only,
44      not evaluated broadly across query types.
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47 MEDIUM PRIORITY – operational maturity, needed soon after launch
48 -----
49 [ ] Hosting for voice_server.py
50     Currently run via `uvicorn --reload` locally. Needs a real always-on
51     host that can reliably reach LiveKit's SIP infrastructure.
52
53 [ ] Basic error visibility / alerting
54     No monitoring currently exists beyond terminal logs. If something
55     breaks in production, there's no way to know until a caller reports it.
56
57 [ ] Concurrency under real multi-caller load
58     Reasoned through architecturally (asyncio.to_thread for blocking
59     search calls, module-level singleton vectorstore/embeddings objects
60     are safe to share) but never actually load-tested with simultaneous
61     real callers.
62
63 [ ] Housekeeping
64     - __pycache__/ folders have been showing up in zip exports; confirm
65       .gitignore covers __pycache__/ and *.pyc before committing.
66     - Unused FrameLogger import still in pipeline.py (flagged, not removed).
67     - loader.py's file_path is a hardcoded absolute Windows path – fine for
68       now, fragile if the project ever moves machines.
69
70
71 KNOWN SEPARATE ISSUE – not voice, do not forget
72 -----
73 [ ] chatbot/chat_server.py is currently BROKEN
74     It imports `from query import build_filter, hybrid_search` expecting
75     data/query.py and data/common.py, both of which were deleted during
76     the voice retrieval rebuild. This will crash on the first /chat
77     request. Chat was never given its own isolated retrieval rebuild the
78     way voice was – needs either a rebuilt retrieval stack (mirroring
79     voice/retrieval/) or restored copies of the deleted files, whichever
80     direction is chosen for chat later. Not blocking voice-only deployment,
81     but will bite whenever chat is revisited.
82
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